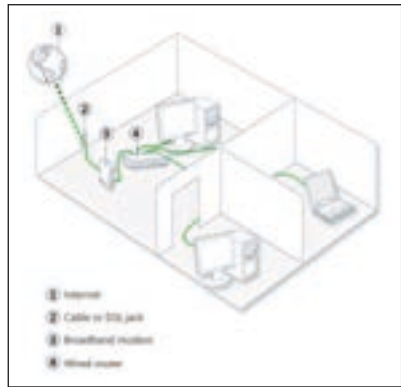


information that they receive – as a result, they can send that information to only the computers that are supposed to receive it. Besides, switches can also send and receive information at the same time; making them much faster than hubs. If your home network has four or more computers, or you want to use your network for activities that require passing plenty of information between computers (such as playing network games or sharing music), you should probably use a switch instead of a hub. Switches cost little more than hubs.



Network with a Wired Router

As with hubs, ethernet implementations of network switches are the most common. Mainstream ethernet network switches support either 10/100 Mbps fast ethernet or gigabit ethernet (10/100/1000) standard. Different models of network switches support differing numbers of connected devices. Most consumer-grade network switches provide either four or eight connections for ethernet devices. Switches can be connected to each other via a so-called daisy chaining method to add progressively larger number of devices to a LAN.

Routers

Routers are devices which enable computers to communicate. They have the capability to pass information between two networks – such as between your home network and the internet. It's this capability to direct network traffic, which gives a router its name. Routers can be wired (using ethernet cables) or wireless. Hubs and switches work well if you just want to connect your computers; however, if you want to give all of your computers access to the internet using one modem, use a router or a modem with a built-in router. Routers also typically provide built-in security, such as a firewall. Understandably, routers are more expensive than hubs and switches.

Home networkers often use an Internet Protocol (IP) wired or wireless router, IP being the most common OSI network layer protocol. An IP router such as a DSL or cable modem broadband router joins the home's local area